The Impact of Screen Time on Academic Performance

Group Members: Lydia Efrem, Mila Brkic, Adrien Rozario, Mira Palakodety, Arjun Sil, Safiyah Fatima

Group Name: Screentime Bots, Group 6

1. Introduction

Many college students have high screen time, and understanding its relationship with academic performance can help students assess and improve their productivity outcomes. Given the increasing digitization of social life and educational tools, understanding how screen time impacts academics is relevant and significant. This topic was chosen because of the sensitive nature of the data involved and the importance of understanding how to handle such data while upholding student's privacy. As such, our group utilized survey data from UMD students to analyze the impact of this relationship to help determine whether students who spend less time on their screen time maintain a higher GPA.

The data collection process focused on data privacy, data anonymity, and data validity. Data privacy was obtained by having informed consent statements and informing respondents why and how their data will be used. Data anonymity was obtained by removing personal identification information: no email was recorded nor names and IDs listed. To ensure data validity, ambiguous wording for questions was avoided, and typed data responses such as GPA entries were cleaned to ensure responses were in the same format.

2. Data Collection

a. Collection Process

The data collection process began with the design of a Google Forms survey. An informed consent statement was included before the survey began to ensure that respondents understood how and why their data would be used. The survey was divided into two sections, one focusing on screen time and the other on academics. The target audience consisted of UMD students from varying academic standings, majors, and graduation dates to get a representative sample of the university population. To ensure this, responses were restricted to terpmail email addresses. The survey was distributed through individual outreach. After collection, the data was cleaned to ensure all responses were complete and correctly formatted for analysis.

In designing the survey, most questions were made required while leaving a few optional. Questions such as average daily screen time, semester GPA, purposefulness of screen time, academic year, and number of credits were required, as they were essential to make meaningful relationships for statistical analysis. While questions regarding which apps students use most frequently and whether they set time limits were optional because they were not essential in the main objective of this data collection.

b. Consent Policy

"Your participation in this survey is completely voluntary and your data will remain confidential. You will not be asked to provide personal identifying information so you will remain anonymous. Your information will be used to analyze the impact of screen time on academic performance. The information you provide will be used solely for academic purposes as part of our DATA 200 course project, which focuses on analyzing the impact of screen time on academic performance. Do you consent to participating in this survey?”

c. Collection Form

https://forms.gle/bJXnYw56pq88MLu4A

3. Collected Data

a. Quality of Data

The survey yielded both quantitative and qualitative data, consisting of ordinal, categorical, continuous, and open-ended responses. The response for phone hours per day is ordinal, app category rank is ordinal, top 5 apps is categorical (nominal), screen time meaningfulness is ordinal (likert scale), phone addiction is ordinal (likert scale), app time limit is categorical (nominal), apps with time limits is categorical (nominal), academic year is categorical (ordinal), expected graduation is categorical (nominal), major type is categorical (nominal), last semester credits is ordinal, semester GPA is continuous, cumulative GPA is continuous, study hours per week is ordinal, and phone distraction during study ordinal. This wide range of data types is compatible for statistical analysis and further interpretation of the relationship between user behaviors and their academic outcomes.

b. Quantity of Data

Before beginning the analysis, a total of 67 survey responses were collected from UMD students by sending them out through personal networks on iMessage and Instagram. It is important to note that this limited the sample population to certain demographics within particular social circles, so the data may not represent the general UMD student population.

c. Response Sheet

https://docs.google.com/spreadsheets/d/1jHgt6Pw2COHhVLtiKH00DaRoV9HKsQLL_dVPkri1V9I/edit?usp=sharing

4. Data Processing

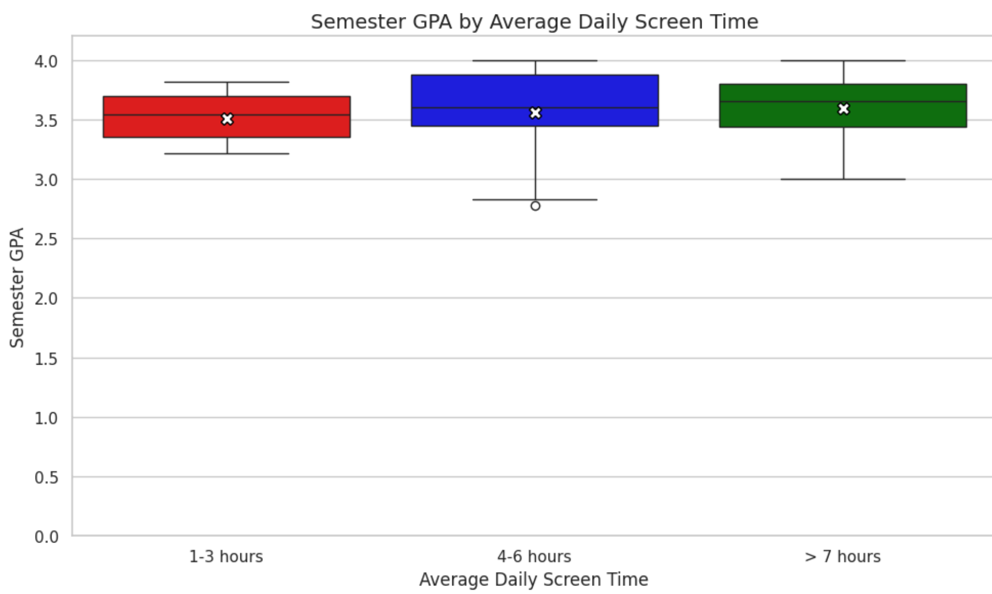
In order to conduct a statistical analysis with the data, no new responses were accepted before generating a spreadsheet of the 67 survey responses from Google Forms. Then, the data was processed by reviewing all survey responses to ensure consistency, checking if GPA data was in a standard numeric format for calculations, and removing missing entries. After ensuring that the data was clean, it was reformatted to make the analysis stages easier. The study times and screen times were mapped as follows: <1 hour mapped to 1, 1-3 hours mapped to 2, 4-6 hours mapped to 3, and >7 hours mapped to 4. Doing so allowed for the generation of regression graphs using Spearman's correlation to analyze linear relationships in the data.

5. Detailed Analysis

Six primary questions were the focus of the detailed analysis. Given the different types of data collected, we used multiple analytical techniques such as Spearman's rank correlation for ordinal relationship, linear regression, and chi-squared tests for categorical relationships.

Analysis 1

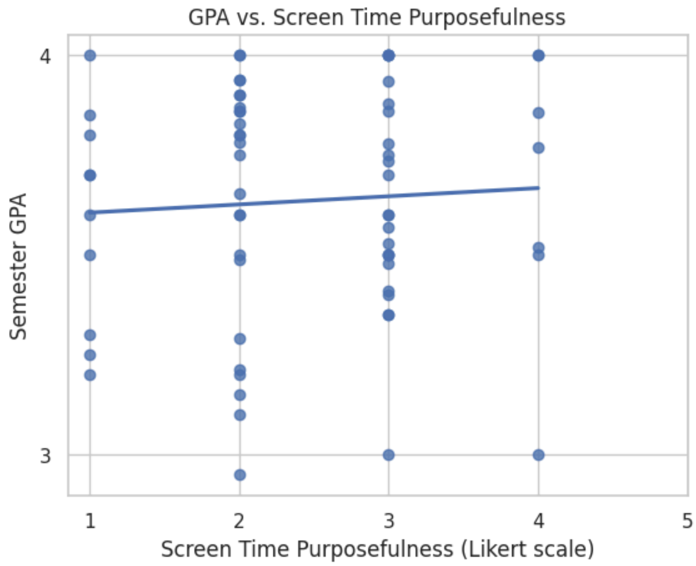
The first question asked was: Does higher screen time correlate with lower GPA? To test this, the relationship between average daily screen time (ordinal) to Semester GPA (numeric) was examined using a Spearman Correlation test. The relationship was visualized through a grouped box plot. The Spearman Correlation value was -0.019 and the p-value obtained was 0.878, which is greater than 0.05, meaning the relationship is not statistically important. Therefore, average screen time was not significantly correlated with semester GPA in this dataset.



Analysis 2

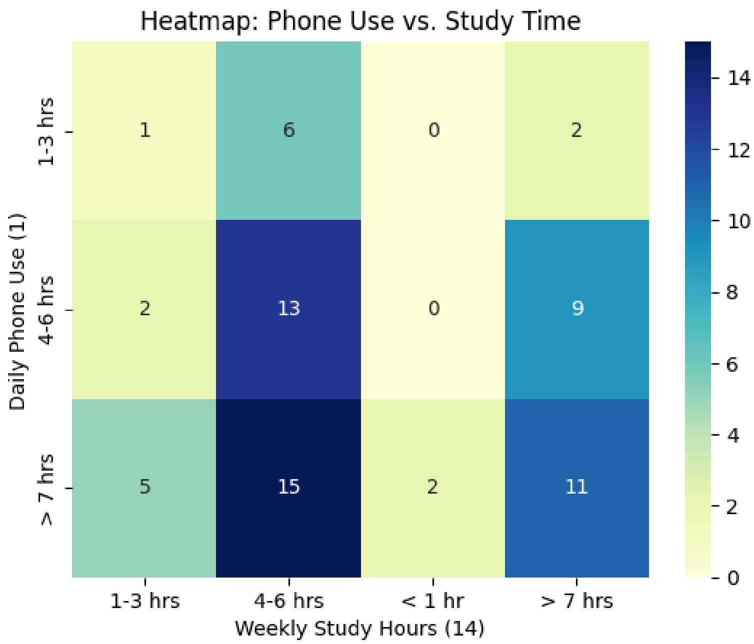
The second question asked: Does meaningful screen time correlate with GPA? To answer this question, a Spearman Correlation test was utilized to determine if there was a statistically significant correlation and visualized the relationship between the two variables using a scatter plot with a regression line. The calculated Spearman Correlation coefficient (the rho) was 0.06, and the p-value was 0.6293. No significant correlation was found, according to the calculations.

However, the scatterplot's regression line shows a slight upward trend, which may hint at a weak positive association. However, this pattern is not strong enough to be conclusive.



Analysis 3

The third question asked was: Do students who study more have lower screen time? To test this, the relationship between average weekly study hours (ordinal) to average daily screen time (ordinal) was examined using a Spearman Correlation. The Spearman Correlation coefficient calculated was 0.06 and the p value obtained was 0.66293 The relationship was visualized through a heatmap for phone use versus study time. Since the p-value is greater than 0.5, the relationship is not statistically important. Therefore, study time was not significantly correlated with average screen time in this dataset.



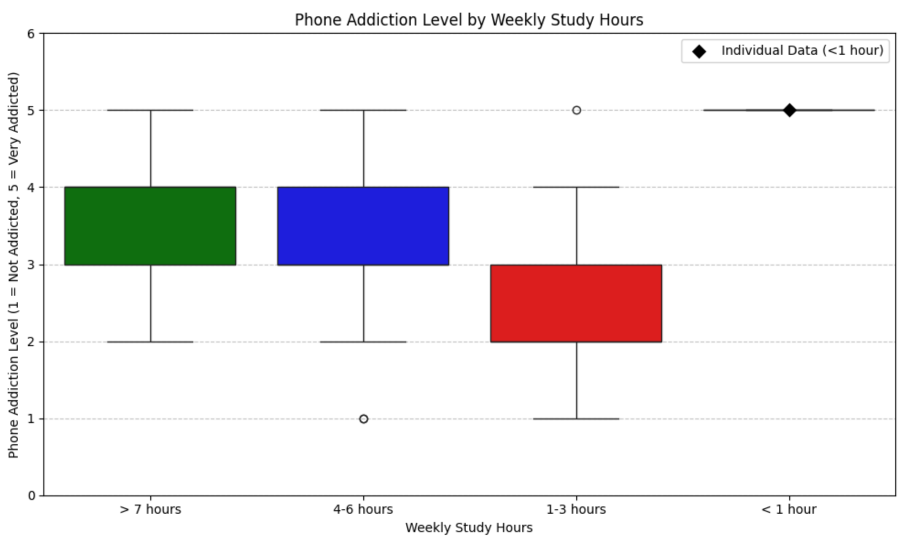
Analysis 4

The fourth question asked was: Do students with more credits have lower screen time? To test this, the relationship between students' reported screen time and credit load during the last semester was analyzed by using a chi-squared test and generating an expected frequencies table for visualization purposes. The most visible pattern demonstrated in the table is shown within students taking 12-15 credits, however most respondents of the survey fall within the 12-15 credit range. This ultimately presents a bias in the data, and therefore cannot be used as a reliable pattern to draw interpretations from. Thus, there is no statistical evidence to support a relationship between credits and screen time.

```
Expected Frequencies Table:
Screen Time  1-3 hours  4-6 hours   >7 hours
Credits
11-13         0.830769   2.123077   3.046154
12-15         4.569231  11.676923  16.753846
<11           0.138462   0.353846   0.507692
>15           3.461538   8.846154  12.692308
```

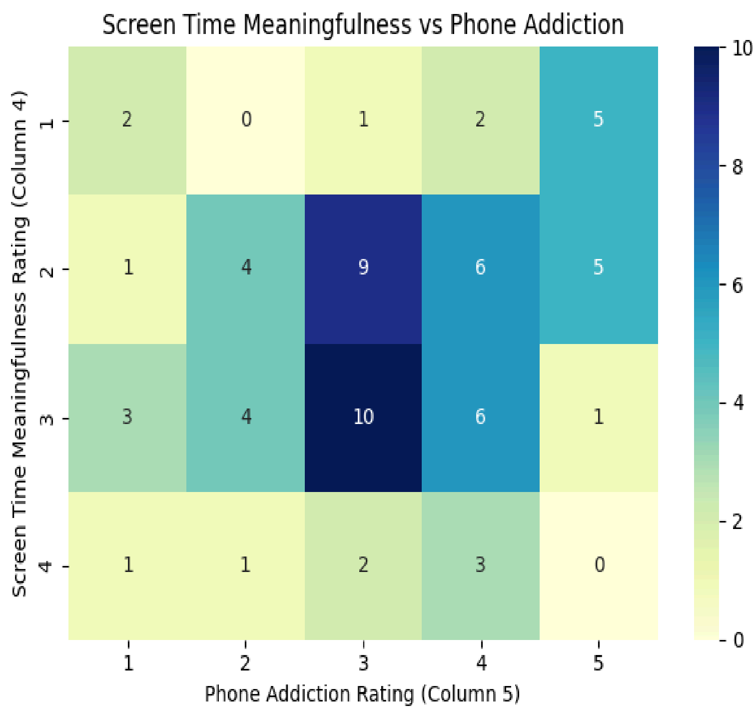
Analysis 5

The fifth question asked was: How does weekly study time (numeric) relate to phone addiction (ordinal)? To find any correlation between these two variables, a Spearman Correlation test was conducted and grouped box plots were made visualizing students' phone "addictions" versus weekly screen times. The Spearman Correlation coefficient was determined to be 0.354 and the p-value obtained was 0.559. Since the p-value is greater than 0.5, the relationship between weekly study time and phone addiction are not statistically significant. The plot obtained shows no meaningful relationship between the two variables.



Analysis 6

The final question asked in this investigation was: Do students who rely less on their phones (numeric) report higher levels of productivity (ordinal)? Once again, a Spearman Correlation test was utilized to see if people who are more "addicted" to their phones exhibit meaningful screen time usage. To visualize any statistical significance, a heatmap was developed to see the relationship between phone addiction and screen time meaningfulness by assessing students' phone reliances vs. GPAs. The Spearman Correlation coefficient was calculated to be about -0.26 and the p-value obtained was approximately 0.03. Since the Spearman coefficient is a negative value and the p-value is less than 0.5, there is enough strong evidence to conclude that a negative relationship exists between these two variables. Thus, as students' phone "addiction" levels increase, their GPAs tend to decrease.

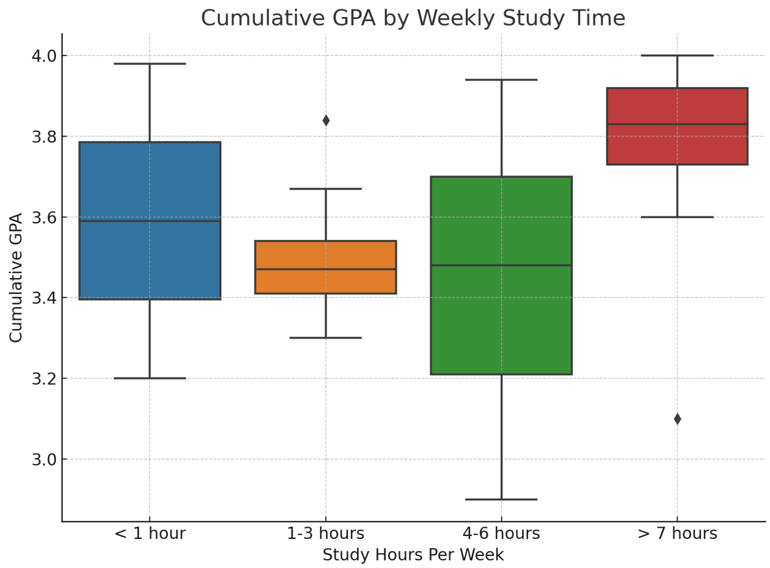


6. Hypothesis Testing

For the hypothesis testing part of our project, we conducted two different tests that checked the correlation between time spent studying and personal thoughts on the meaningfulness of their screen time against each respondent's GPA.

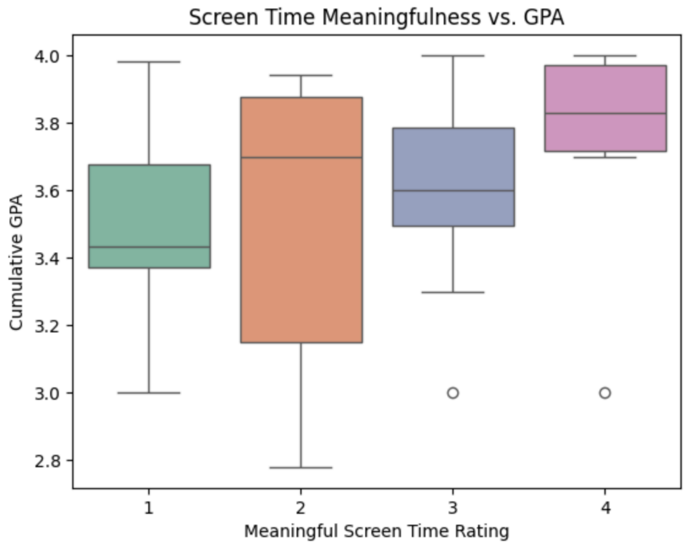
Hypothesis Test 1

The first test was done with the Kruskal-Wallis method, which tests to see if the medians between two independent variables are statistically significant. For this test, the question we wanted to answer was, “Does the amount of time spent studying correlate with higher academic performance?” The null hypothesis stated that there was no significant difference in GPA across study time groups, and the alternative hypothesis stated that at least one study time group had a cumulative GPA that statistically differed from the other groups. After conducting this hypothesis test, we found that the p-value was 0.00038, which is less than 0.05, meaning we rejected the null hypothesis. From that and by viewing the boxplot showing the correlation between study time and cumulative GPA, we can conclude that there is a significant correlation between study time and GPA. Respondents that reported studying over 7 hours a week had a median cumulative GPA of about 3.85, which is much higher than the medians of the other groups. Interestingly, the 1-3 hours and 4-6 hours groups had about the same median GPA, while the less than 1 hour group had a slightly higher median GPA. Additionally, there were two outliers: one person who studied more than 7 hours a week but had a 3.1 GPA and another person who studied 1-3 hours a week but had about a 3.8 GPA.



Hypothesis Test 2

The next hypothesis test was done with the Spearman Correlation method, which checks if there is a significant increasing or decreasing relationship between two independent variables. For this test, the question we wanted to answer was, “Does rating screen time as meaningful correlate with higher academic performance?” The null hypothesis stated that there is no correlation between screen time meaningfulness and GPA, and the alternative hypothesis stated that there is a positive correlation between meaningfulness of screen time and GPA. After conducting this hypothesis test, we found that the p-value is 0.103, which is greater than 0.05, and Spearman’s rho is 0.204, which is greater than 0. Since the p-value is greater than 0.05, we failed to reject the null hypothesis, meaning there is no statistically significant correlation between reported meaningfulness of screen time and GPA. Although there is no statistically significant correlation, we can look at the boxplot and see that the respondents who ranked the meaningfulness of their screen time as a 4 out of 5 on a likert scale tended to have higher GPAs. But other than that, there is not a statistically significant difference between the ranking of screen time meaningfulness and respondents' GPAs.



7. Future Steps

For the future steps of our project, there are a variety of things we can do. One aspect of our survey that could've possibly led to failing to find correlation between our variables is the fact that we didn't have a lot of respondents. We had a total of 67 respondents for our survey, however when we updated the survey to have more questions and we sent it out again, we only had an additional 35 respondents who answered those questions. This means that our questions had either 67 or 35 respondents, depending on if the question was a part of the survey before or after we updated it. One thing we could do is recreate the survey with all of the questions that we intended to ask, possibly adding more, and then sending it out again so all of our questions had the same number of responses, making it easier to compare independent variables. By sending out the survey again, we could also get many more respondents. Although 67 is not a small amount, more would allow us to draw more accurate conclusions with statistical analysis. Additionally, we had mostly STEM majors respond to our survey after we updated it, with 29 respondents saying they're pursuing a Bachelor of Science and 6 saying they're pursuing a Bachelor of Arts. STEM and humanities majors arguably have different standards when it comes to academics (exams vs papers, etc.), so if we had more people pursuing a B.A. respond, our results could possibly look different.

The first change in a new survey would be prioritizing numerical data instead of ordinal or categorical data. It would be easier to conduct statistical analysis on variables that are both purely numerical. Categorical data is often left to be analyzed on individual cases, comparing the answer to the rest of the answers of the respondent, which is more complicated compared to methods done with numerical data. To avoid this, we would ask for exact numbers instead of giving students fixed time ranges to choose from. This would allow us to treat the data as continuous, which opens up more advanced and accurate types of analysis, such as linear regression and correlation tests that assume numeric input.

Another change that could be implemented is to ask about screen time on other devices. Many people use laptops to do most of their work as opposed to their phones or tablets. This would open up possibilities of analyses of screen time across devices in relation to GPA.

8. Conclusion

This project studied the relationship between screen time and academic performance in college students. Unlike other studies that focus only on social media, our project looked at broader phone use. We asked students about how meaningful they felt their screen time was, whether they used app limits, and what types of apps they used most often. This helped us understand that habits and personal attitudes toward phone use may have a stronger connection to academic performance than simple usage time. Our results support the idea that it is not just how long students use their phones, but how and why they use them that matters.

While many people assume that more time spent on phones leads to lower grades, our results suggest the issue is more complex. Most of the relationships we tested were not statistically significant. The amount of time students spent on their phones did not have a clear impact on GPA. However, one result stood out. We discovered that students who felt more dependent on their phones tended to report lower GPAs. This suggests that the feeling of phone reliance may be more important than screen time itself.

This study can help students become more aware of their own habits. By understanding how their phone use may affect their academic performance, they can begin to make changes that support their success. Though we cannot scientifically determine that all screen time is harmful, we can confidently conclude that mindful phone use is a step toward better academic outcomes and healthier study habits.